

NexGen iMATCH System Architecture

The as-built forensic face recognition platform,
and the architecture it is being built toward

This document is the complete architectural record of NexGen iMATCH, a forensic 1:1 verification and 1:N identification platform for investigative use. It is written in two parts. **Part A** describes the system as built and running: the process model, every layer from evidence intake to court report, the recognition engine internals, the persistence and security design, and the measured operating envelope. **Part B** describes the target architecture — the layered evidence-first design that the current system is being migrated toward, with each layer marked built, partial, or design. Every figure quoted here was produced on the project's own hardware against the project's own corpora; nothing is carried in from literature.

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PROJECT	NexGen iMATCH — forensic face recognition and identity intelligence
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SECTION 00

Document control and provenance

This is one of two companion volumes recording the design and construction of NexGen iMATCH. It covers architecture. The second volume covers the implementation plan and the order in which the work was carried out.

Author	Lochan Sharma. Sole architect, implementer, and author. All design decisions, measurement protocols, code and documentation in this project are her own work.
Period	June 2026 through 5 August 2026 — approximately two months. Design, dataset acquisition and the initial codebase were carried out in June, before version control began; the repository's first commit lands on 1 July 2026 and carries that earlier work in as its opening 75 files.
Version control	103 commits across 18 tracked working sessions. The heaviest single day, 31 July 2026, ran from 01:30 to 23:30 across 39 commits.
Evidence base	This document is derived from the repository itself: the computed import graph from the service entry point, the FastAPI route table, the SQLAlchemy schema, the benchmark record, and the experiment result files. It is not written from recollection.
Reproducibility	Every quantitative claim in this document traces to a named artefact in the repository. Where a figure is directional rather than final, it is labelled as such.

103	351	102	56	36	9
COMMITTS	TESTS PASSING	PYTHON MODULES	IN THE REQUEST PATH	HTTP ENDPOINTS	PERSISTED TABLES

HOW TO READ THE STATUS MARKINGS

Every component in this document carries one of four markings. **Built** means it is in the request path and covered by tests. **Instrumentation** means it runs from the command line and produces a measurement artefact, but is not in the request path. **Partial** means the code exists and is exercised, but a stated dependency is unmet. **Design** means it is specified and not yet implemented. The presence of a file is never on its own a claim that a capability is delivered — for a forensic system that is precisely the wrong inference to invite.

SECTION 01

Architectural thesis

Face recognition on clean imagery is a solved problem, and forensic casework is not clean imagery. The architecture of this system is shaped entirely by that gap and by a second observation about what a forensic user actually needs from a match.

The measured gap

The same pipeline, the same weights, the same arithmetic, measured across corpora of descending quality, produces the following. These are this project's own numbers, not published ones.

Corpus	Character	Measured result	Reading
LFW	Clean, frontal, web-collected	99.70% verification accuracy	Saturated. Useful only as a regression detector.
AgeDB-30	Clean, cross-age	96.03% at the calibrated point	Age is a real but bounded cost.
CPLFW	Clean, cross-pose	Cllr 0.2616	Pose costs more than age.

Corpus	Character	Measured result	Reading
TinyFace	Native low-resolution, real	TAR@FAR=0.1% of 33.13% (R50); 60.77% (ViT-B)	The condition the product exists for.
QMUL-SurvFace	Native surveillance capture	TAR@FAR=0.1% of 8.8% at arm A	Where the honest limit lives.
SCface (D1, 4.2 m)	Real optics at real distance	TAR@FAR=0.1% below 1%	Real optics is harder than synthetic degradation.

Figure 1.1 — the quality ladder. A decade of architectural progress moved the top row by three orders of magnitude and the bottom row by fractions. That asymmetry is the signature of a formulation problem, not a compute problem.

The second observation: a matcher answers the wrong question

A 1:N matcher returns the nearest neighbour. That question always has an answer. For a probe whose subject is not enrolled at all, the rank-1 candidate is structurally indistinguishable from a true hit — this is definitional, not an implementation defect, and every deployed identification system has it. A forensic user does not need to know who is closest. They need to know how much the evidence weighs, against what population, with what uncertainty, and whether the honest answer is that the imagery cannot support a conclusion at all.

THE THESIS IN ONE SENTENCE

The architecture separates **recognition** from **evidence**: the recogniser produces a score, and a distinct evidence layer converts that score into a calibrated evidential weight against a declared reference population, with an explicit not-enrolled hypothesis and the option to abstain. Every structural decision in this document follows from that separation.

SECTION 02

Standing design rules

These nine rules are carried forward across every phase of the project. They constrain the architecture more than any performance target does.

#	Rule	Why it is a rule and not a preference
1	A capability is claimed only where a test or a measurement backs it	The claims register is a tracked document and is re-run after any model, threshold or fusion change. Wording is corrected to match the evidence, not the other way round.
2	No silent fallback to a degraded path	An earlier revision had an 'auto' engine mode that fell back to hashing pixels into a vector when weights were missing. The API kept answering and every score was meaningless. The engine now either loads real weights or raises.
3	Contamination figures are reported as floors, never as totals	An overlap audit finds what it can detect. Reporting a detected overlap as the total overlap understates the risk in the direction that flatters the result.
4	Negative results are published, not buried	The project's own central research thesis was refuted on real optics and the refutation is a first-class artefact. A programme that only publishes its wins cannot be trusted on its wins.
5	Gates are pre-registered before the run	Every experiment fixes its pass and fail thresholds in writing before the data is seen, so a marginal result cannot be tuned into a pass.
6	Generative output never enters the evidential path	Enhanced or restored imagery carries a distinct type that the matching and likelihood code structurally cannot accept. This is enforced by the type system, not by convention.
7	Tenant isolation is structural, not a filter	Each tenant's vectors live in a separate matrix and the search function takes the tenant as a required argument. A post-hoc filter over a global search is one forgotten predicate away from a cross-tenant biometric leak.
8	Thresholds are model-specific and never carried across packs	The operating point is a property of a specific set of weights measured on a specific corpus. Copying it to another pack silently invalidates the false-match control.
9	The audit record is written before the response is returned	An audit entry written after the fact is an audit entry that can be lost on the failure path that most needs it.

PART A

The system as built

Everything in Part A is running code. Each section states what is in the request path, what runs as offline instrumentation, and where the boundary between them sits.

SECTION 03

Process model and topology

The system is a single-origin stack. The browser talks to one origin; that origin serves the static application and proxies the API. This is deliberate: the split-origin arrangement that preceded it produced four stacked authentication failures — cookie domain, CORS credentials, CSRF token origin, and refresh-token path — that presented to the user as a single unhelpful 'Failed to fetch' on login.

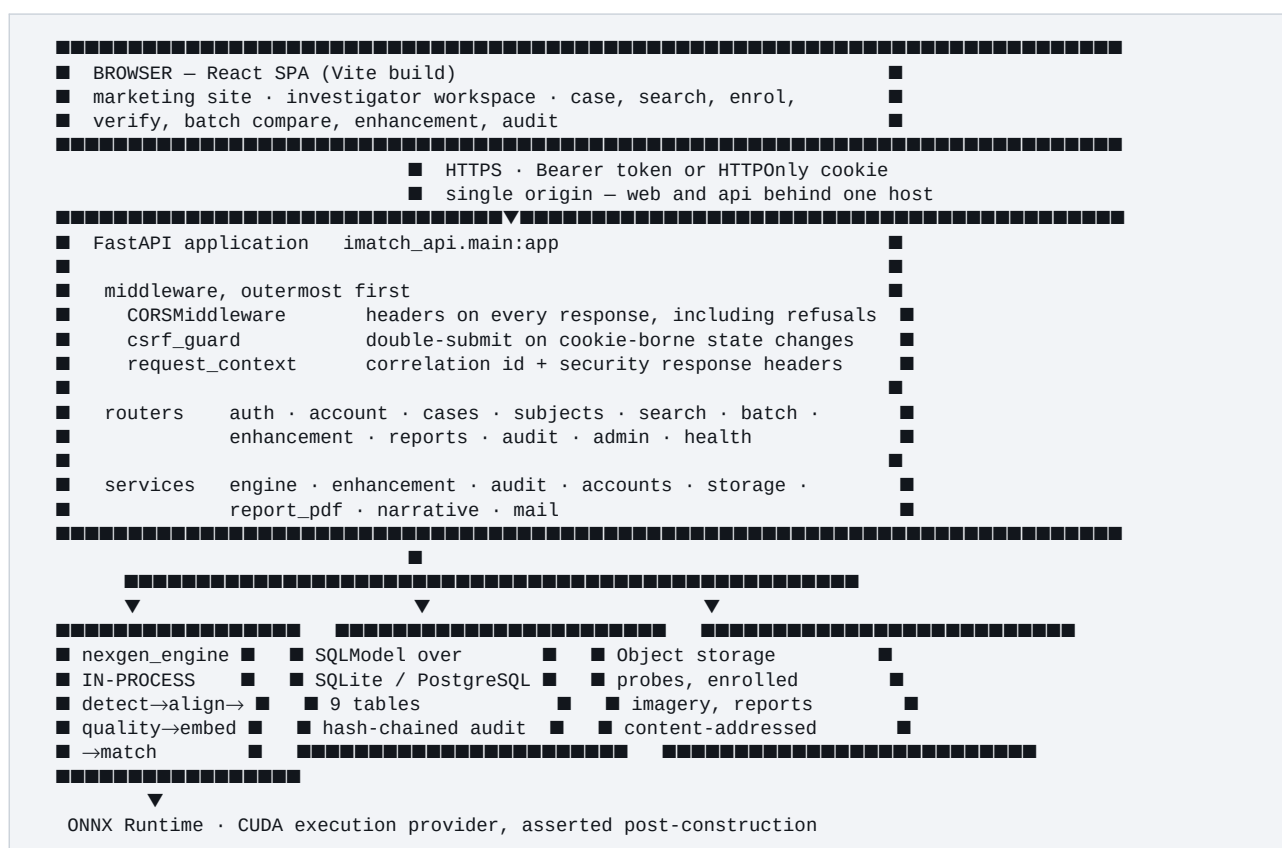


Figure 3.1 — process model. The recognition engine is in-process by design; there is no model server. A request thread calls the inference session directly.

Why the engine is in-process

A separate model server buys horizontal scaling and costs a serialisation hop on every probe. At the measured throughput of this system the hop is a material fraction of the request, and the scaling it buys is not yet needed. The consequence is recorded honestly: threading saturates at about four workers, so **batching, not more workers, is the scaling lever**. That is a bound the architecture accepts rather than hides.

Deployment topology

Target	Shape	Notes
Local development	Single process, SQLite, CPU or CUDA	The primary working configuration. Owner session persists locally so the workspace is reachable without an external mail round-trip.
Self-hosted stack	Docker Compose: web · api · postgres, with named volumes for model cache, API data and the database	Single origin. The intended production shape for an on-premises forensic laboratory, where the imagery must not leave the building.
Managed cloud	Blueprint retained, not the plan of record	Split-origin hosting was tried and abandoned: it forced the biometric payload across a network boundary for no operational gain, and produced the login failure class described above.

TRUST BOUNDARY

The engine, the gallery and the audit chain all sit inside one process on one host. The only boundary the biometric data crosses is browser-to-origin. For a forensic deployment this is a feature: the smallest possible number of places where a probe image exists.

SECTION 04

Layer map

Eight layers, from the arrival of an image to the production of a court-usable report. The status column is the honest state of each layer today.

Layer	Responsibility	Principal modules	Status
L0 · Evidence intake	Decode, geometry guard, content hashing, storage, provenance record	storage_service · forensics/provenance	Built
L1 · Sensor and degradation	Blind PSF, noise and JPEG estimation; MTF passband; band-limit projection	degradation/{psf,noise,jpeg,bandlimit, estimate}	Built as instrumentation
L2 · Representation	Detect, align, quality-gate, embed to a comparable template	detection/* · inference/pipeline · models/*	Built
L3 · Comparison	Cosine similarity, score fusion, condition removal, decision banding	inference/score_fusion · config thresholds	Built
L4 · Evidence engine	Cllr, calibration, likelihood ratio, information bounds, open-set, population	forensics/{metrics,calibration,information,openset,population,evidence}	Partial — metrics and calibration validated; population needs identity-labelled cohorts
L5 · Memory	Case, subject, template, search and candidate records; replayable evidence log	db/models · forensics/memory	Built for the relational model; replayable posterior is design
L6 · Enhancement	Two-track investigator-facing image improvement, held outside the evidential path	enhancement/{analysis,planner,runner ,registry,backends}	Built, gated
L7 · Reporting	Examiner-grade PDF, narrative, audit excerpt, stated limitations	report_pdf · report_service · narrative_service	Built

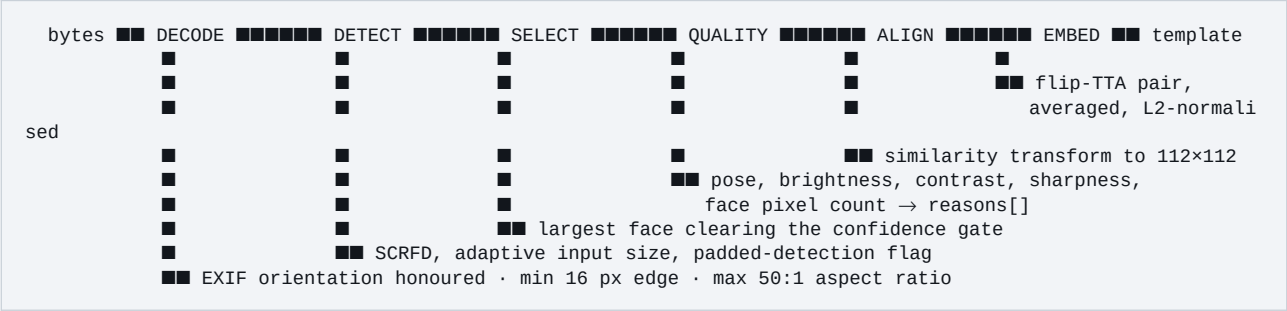
THE LAYER THAT CARRIES THE DIFFERENTIATION

L4 is the layer nobody else is competing on. Matching accuracy is contested by organisations with proprietary data and decades of investment. Evidential weight against a declared population, with an explicit not-enrolled hypothesis and a stated information bound, is unclaimed — and it is the thing a court actually needs.

SECTION 05

The recognition engine

Image bytes in, comparable biometric template out. Six stages, each of which exists because a specific failure was observed without it.



Decode — the geometry guard

EXIF orientation is honoured, because phone photographs routinely carry a rotation flag and ignoring it feeds the detector a sideways face and silently collapses the match rate. Two geometric rejections follow, and both were added in response to real faults rather than anticipated ones: the detector scales an image to fit its input while preserving aspect ratio, so a 4000×1 strip rounds to zero height and raises from deep inside the vision library as an unmapped 500. A 16 px floor on each edge and a 50:1 aspect ceiling convert that into a typed, readable rejection — which in a batch is the difference between one skipped item and a dead request.

Detect and select

Detection uses SCRFD with an adaptive input size: the configured 640×640 is a ceiling, and the detector picks the smallest size that fits each image, so it is not a fixed cost. Face selection takes the largest candidate that clears the confidence gate. Where more than one face is present the result carries a `multiple_faces_detected` reason — the system does not silently pick one and present it as the subject.

Quality gating

The quality filter is a gate, not a score to be optimised. It evaluates detection confidence, face pixel count, yaw, pitch, roll, brightness, contrast and sharpness, and emits a list of reasons rather than a single number. Any reason present sets `review_required`. The gate defaults are deliberately conservative:

Gate	Default	Rationale
Minimum quality score	0.35	Composite floor below which the template is not treated as decision-grade.
Minimum detection confidence	0.70	Higher than the detector's own 0.50 floor, which is used for enumeration rather than selection.
Minimum face pixels	60	Below this the reflectance component of identity is absent rather than degraded.
Maximum yaw / pitch / roll	35° / 30° / 25°	Pose is the largest single clean-imagery cost measured in this project.
Brightness window	40 – 220	Clipped highlights and crushed shadows both destroy the geometry the recogniser depends on.
Contrast / sharpness floor	0.30 / 0.30	Separates genuinely unusable capture from merely poor capture.

Embed — and the two things the pipeline deliberately does not do

The template is the model output averaged over the aligned crop and its mirror, then re-normalised. Horizontal flip is the only augmentation applied, because a face and its mirror should encode to the same identity so averaging cancels part of the

pose-specific noise; brightness and sharpening shifts instead move the crop off the manifold the recogniser was trained on and cost accuracy. Both crops go through in a single batch.

- **It does not fuse unrelated backbones through a random projection.** A projection never trained jointly with an encoder destroys the metric structure the encoder learned. An early revision did exactly this and the resulting 'ensemble' was worse than its own best member.
- **It does not adjust a stored template using statistics from previous queries.** Cohort normalisation belongs at the score level, where it cannot make a template depend on search history. A template that changes because of what was searched before it is not evidence.

DETERMINISM

The same image always produces the same template, regardless of what was searched before it, what else was in the batch, or which execution provider is bound. CUDA and CPU run the same model and the same arithmetic; the device changes latency, never the answer.

SECTION 06

Model layer

Three model roles, each with an explicit status and an explicit reason for being there.

Role	Model	Status	Evidence for the choice
Detection	SCRFD (buffalo_1 pack)	Production, frozen	Measured and adequate; detection is not the bottleneck. Detection cost was brought from 939 ms to 54 ms during the rebuild by removing a redundant re-detection pass.
Recognition	ViT-B KP-RPE AdaFace	Production since 3 Aug 2026	Replaced the ArcFace R50 on a single-variable comparison: TinyFace TAR@FAR=0.1% moved 20.59% → 60.77%, a 3.0× gain, against a 153,428-image reference population and a 50-million-pair impostor distribution.
Proposal / retrieval	ArcFace w600k_r50	Retained, demoted	Fast, well-characterised, and the basis of every calibration measurement in the project. It is now an initialiser and a retrieval front-end, not the identity representation.
Ensemble (3-model)	InsightFace multi-pack	Rejected	The shipped three-model ensemble lost to the best single model on four of five datasets at three times the compute. Both candidate architecture diagrams called for a six-model ensemble; the project's own data rejected it.
LoRA fine-tune	Adapter over the production backbone	Investigated, not shipped	QMUL TAR@FAR=0.1% moved 2.31% → 22.35% and passed two of three gates. The third — condition leakage — failed, and SCface then refuted the synthetic result on real optics. Documented and not deployed.

Weights governance

- Every weight file is pinned by SHA-256 checksum, verified at load, and recorded in the model card.
- The backbone is loaded only on a cache miss, so a warm process does not pay the load cost per request.
- Two silent integration defects were found and fixed during the backbone swap: a channel-order mismatch (BGR against RGB) that degraded scores without raising anything, and a normalisation difference between packs. Both produced plausible-looking numbers, which is what makes that class of defect dangerous.
- The engine asserts on the *post-construction* execution providers of a real inference session. The build-time provider list is not trustworthy: it reports providers the wheel was compiled with, not providers that actually loaded. This project lost its GPU configuration twice to that exact false positive.

SECTION 07

Decision layer and operating point

The threshold is the single most consequential number in a forensic matcher, and it is calibrated for false-match control rather than for peak accuracy.

The demonstrated failure that forced the change

The system originally used 0.20, the ten-fold cross-validated accuracy optimum. At that point two different people scored 0.2405 and the comparison was reported as supporting the same person. That is not a hypothetical: it happened, in this system, on real imagery.

Operating point	False match rate	False non-match rate	Selected
0.2000 — accuracy optimum	1.19% (about one false match per 84 impostor comparisons)	3.30%	No
0.2871 — FMR = 0.1% point	0.10%	6.32%	Yes

Figure 7.1 — measured over 40,098 AgeDB pairs for the w600k_r50 pack. A twelve-fold reduction in false matches for roughly double the miss rate.

The asymmetry is the whole argument. A missed lead costs a re-check. A false match points investigators at an innocent person. For forensic use the objective is not accuracy, and the decision record states that explicitly at the point of definition in the code, so that anyone tempted to 'optimise' the number reads the reasoning first.

Review banding

A review threshold sits at 0.75 of the match threshold, so near-miss candidates still reach a human rather than being silently discarded. This ratio is a carried-forward convention and is labelled as such: it has not been separately optimised, and the document says so rather than presenting it as a measured value.

THRESHOLD PORTABILITY

The operating point is a property of one specific set of weights measured on one specific corpus. It is re-derived after *any* change to the model pack or the embedding pipeline, and it is never copied to another pack. Adopting the ViT backbone required a full recalibration, not a carried-over number — and that requirement is recorded as a mandatory consequence in the decision record rather than left to be remembered.

SECTION 08

Search and gallery index

An in-memory vector gallery, partitioned by tenant, searched exhaustively.

Tenant isolation

Each tenant's vectors live in a separate matrix, and the search function takes the tenant identifier as a required argument. There is no code path that can compare a probe against another tenant's templates. The alternative — a global search with a tenant filter applied afterwards — is one forgotten predicate away from a cross-tenant biometric leak, and the difference in cost between the two designs is nil.

Exact search, and the decision not to approximate

Cosine similarity over L2-normalised templates reduces to an inner product, so a brute-force matrix multiply is exact. The measured cost:

Gallery size	Search p50	As a fraction of the ~14.7 ms probe encode
1,000 templates	0.207 ms	1.4% — free
10,000 templates	1.087 ms	7.4% — free
100,000 templates	15.981 ms	109% — the point at which it stops being free

Approximate nearest-neighbour indexing was measured and rejected for this task. The reasoning is stated in full because it is easy to get backwards: an exact flat index would win a constant factor from better vector instructions, not a change in complexity. Genuine scaling past 10■ requires a genuinely approximate structure, and that carries recall loss. **A missed lead is invisible to the examiner** — nothing in the interface distinguishes 'not in the gallery' from 'in the gallery and not retrieved'. That recall loss must therefore be measured and explicitly accepted, not assumed away, and the trigger for adopting it is a measured gallery size rather than a principle.

Subject collapsing and the runner-up margin

A subject enrolled from several photographs appears once in the candidate list, at its best-scoring template; otherwise a well-enrolled subject fills the list and hides every other lead. The search returns not only the ranked candidates but the *full score vector* over the searched gallery, because the decision layer needs it to compute the runner-up margin and the impostor distribution. Without that vector a top score cannot be told apart from a near-tie, and a near-tie is exactly the case an examiner must be told about.

SECTION 09

Persistence and data model

Nine tables under SQLModel, portable between SQLite for single-host work and PostgreSQL for the self-hosted stack.

Table	Holds	Notable fields
Tenant	The isolation unit	slug, active
User	Operator accounts	role, badge number, e-mail verification state, OTP hash and window, reset token hash, failed-login counter, lock expiry, refresh token hash, session epoch
ApiKey	Machine principals	prefix, key hash, role, expiry, last used
Case	The investigative container	reference, status, lawful basis , owner
Subject	An enrolled person	external reference, case linkage, enrolled-by
Template	A biometric template	nonce + ciphertext (never plaintext), dimensions, recogniser backend and pack, quality score, detector, image SHA-256
SearchRun	One search, fully reconstructable	mode, lawful basis, purpose, decision, top score, margin, gallery size, quality, liveness, deepfake risk, review flag, recogniser backend and pack, both thresholds in force , probe SHA-256, duration, audit hash
Candidate	One returned lead	rank, score, normalised score, adjudication, adjudicated-by and -at, examiner notes
AuditRecord	The tamper-evident chain	sequence, actor, action, resource, outcome, lawful basis, IP, user agent, previous hash, entry hash

Three properties worth calling out

- **Templates are encrypted at rest.** The table stores a nonce and ciphertext. A biometric template is not a hash — it is recoverable identity data — and a database dump must not be an identity leak.
- **A SearchRun records the configuration it ran under**, not merely its result: the recogniser backend, the model pack, and both thresholds in force at the time. A result whose operating point cannot be reconstructed is not re-examinable, and a forensic result that cannot be re-examined is not evidence.

- **The audit chain is hash-linked.** Each entry carries the previous entry's hash, so any retrospective edit breaks the chain from that point forward. A verification endpoint walks and checks it. An early defect in this chain — non-deterministic ordering that could report false tampering — was found and fixed; a tamper-evidence mechanism that cries wolf is worse than none.

SECTION 10

API surface

Thirty-six endpoints across eleven routers. The middleware order is load-bearing.

CORSMiddleware	outermost — headers must be present on refusals too, or the browser reports a network error instead of the real 401/403
▼	
csrf_guard	double-submit validation on cookie-borne state changes only; bearer-token calls are exempt because they are not ambient
▼	
request_context	correlation id, security response headers, request timing
▼	
route handler	authentication → rate limit → lawful basis → work

Endpoint families

Family	Endpoints	Purpose
Authentication	login · logout · refresh · me · register · verify-email · resend-otp · forgot-password · reset-password · csrf · users	Full account lifecycle including e-mail verification, password reset and session revocation by epoch.
Cases	list · create · get · update · report	The investigative container and its exported report.
Subjects	list · enrol · get · delete · templates · delete template	Gallery management, with per-template deletion.
Matching	verify · search · batch · searches · candidates · adjudicate	1:1, 1:N, and batch. Adjudication is a first-class recorded act, not a UI state.
Engine	status · metrics	Which execution provider is bound, which pack is loaded, per-stage timings.
Audit	list · verify	Read the chain; verify its integrity.
Admin	api-keys list · create · revoke	Machine principal management.
Health	health	Liveness for the container orchestrator.

Request path — 1:1 verification

1. Two images and a lawful basis arrive at the verify endpoint.
2. Authentication resolves a principal from the bearer token, the session cookie, or an API key.
3. The rate limit is checked against that principal.
4. Lawful basis is required and the request is refused without one when enforcement is on.
5. Each image: decode → geometry guard → detect → align → quality → embed.
6. Cosine similarity is compared against the configured threshold and banded.
7. **The audit record is written and hash-chained before the response is returned.**

The 1:N path is identical through step 5, then compares the probe against the tenant's gallery shard by exact inner product, ranks, collapses by subject, and truncates to the requested candidate count. Batch mode supports both one-reference-against-many-probes and pairwise modes, with within-batch probe deduplication that produced an eleven-fold reduction in GPU work on realistic inputs.

SECTION 11

Security architecture

Control	Implementation	Failure it prevents
Authentication	Password login with hashed credentials; HTTPOnly session cookie or bearer token; API keys with prefix-and-hash storage	Credential recovery from a database dump.
Session revocation	Per-user session epoch; incrementing it invalidates every outstanding token for that user	A stolen refresh token outliving the response to the theft.
Brute-force control	Failed-login counter with a lock expiry	Online password guessing.
E-mail verification	OTP with hash storage, expiry, attempt cap and a send-rate window	Both account-takeover-by-registration and using the OTP endpoint as an outbound mail relay.
CSRF	Double-submit token, enforced only on cookie-borne state changes	Cross-origin state change by an authenticated browser.
Rate limiting	Per-principal, applied before any engine work	Resource exhaustion by an authenticated but abusive principal.
Response headers	Content security policy, frame options, referrer policy, no-sniff, applied by middleware to every response	Clickjacking and content-type confusion.
Template encryption	Nonce and ciphertext at rest	A database compromise becoming a biometric compromise.
Tenant isolation	Separate vector matrices, tenant required at the call site	Cross-tenant biometric leakage.
Lawful basis	Required field on cases and searches, refused when absent under enforcement	A search with no recorded legal justification — the finding that ends a case in court.
Audit chain	SHA-256 hash-linked entries with an integrity verification endpoint	Silent retrospective edit of the record of who searched for whom.
Adversarial input	Typed rejections for degenerate geometry, empty payloads, and undecodable bytes	Unmapped exceptions surfacing as 500s and killing batch requests.

WHAT IS DELIBERATELY NOT CLAIMED

The system reports a liveness signal and a synthetic-media risk score. Both are passive single-image heuristics and both are marked `certified: false`. **Presentation-attack detection is not claimed.** A module implementing it exists in the tree and is deliberately not wired into the service — its being unreachable is consistent with the claim, not an oversight, and the architecture record says so explicitly so that a reader cannot infer the capability from the filename.

SECTION 12

Forensic evidence layer

This is the layer that distinguishes the system from a matcher. It converts a similarity score into evidential weight.

Component	Function	Status
metrics	C_{llr} , C_{llr}^{min} , C_{llr}^{cal} , PAV, Tippett plots	Built and validated
calibration	Condition-conditional score $\rightarrow \log \blacksquare$ likelihood ratio, fitted per quality bin	Built
information	Identity bits, Fano bounds, maximum defensible gallery size	Built; requires an identity-labelled reference population to emit

Component	Function	Status
openset	Not-enrolled hypothesis, conformal candidate sets with coverage	In progress
population	Typicality — the likelihood-ratio denominator and the random match probability	Design; the critical path
evidence	Evidence report with the ENFSI verbal scale and explicit stated limits	Specified
provenance	Content-addressed lineage of what the model was shown	Built
memory	Replayable evidence log; posterior as a pure function of the log	Design

What the measurements say about where the loss is

Two-parameter logistic calibration on held-out folds leaves the calibration component of the cost below 0.022 on every corpus tested. On degraded imagery the total cost is 0.72 — of which **98.7% is discrimination loss, not calibration loss**. This is the most consequential number the project holds, and it points in an uncomfortable direction: the evidence layer cannot rescue surveillance imagery. Only better recognition can. Reporting honestly about a weak signal is necessary and it is not sufficient.

Why the denominator is measured rather than modelled

The likelihood-ratio denominator is estimated empirically against a declared reference cohort, not generated by a model. This is a deliberate risk reduction, and it was learned the hard way: the capacity measurement was broken twice. The first attempt censored 99.7% of its results because three thousand impostor pairs cannot resolve a tail below one in three thousand. The second attempt, with a twenty-million-pair sampled pool, was invalid because the assumption that a different pair implies a different identity is false — contamination tracks identity count, and at 0.2% it was *twice* the false-accept rate being measured. It made one corpus appear to fall from 96.03% to 8.40%.

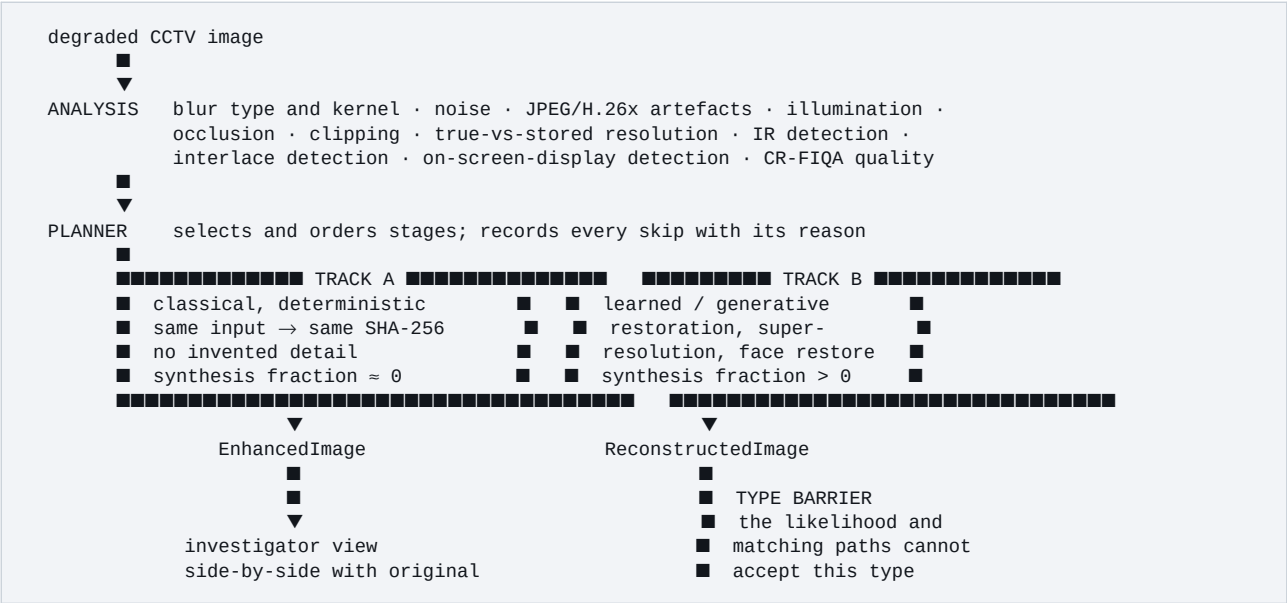
THE LESSON THAT BECAME AN ARCHITECTURAL RULE

Reference-population purity is the measurement, not a caveat on it. The code now refuses to emit a capacity figure without identity-labelled cohorts, rather than emitting a number with a footnote. A number with a footnote gets quoted without the footnote.

SECTION 13

Image enhancement subsystem

A two-track pre-processing layer for investigator-facing image improvement, held rigorously outside the evidential path.



The type barrier

Generatively restored imagery carries a distinct type that the matching and likelihood code structurally cannot accept. This is not a convention or a review checklist item — it is a property the type system enforces, and the report-drawing function that would render an unpaired enhanced image raises rather than rendering. The reasoning is that super-resolution is one-to-many: it picks a plausible mode from a learned prior, so matching against its output is matching against the generator, not against the evidence.

Measuring synthesis rather than disclaiming it

A disclaimer is not a measurement. Four independent estimators quantify how much of the output was invented rather than recovered:

Estimator	Applies to	What it measures
Band-limit synthesis mask	All output	Energy appearing outside the spatial-frequency band the imaging chain physically passed. Detail above the passband did not come from the sensor.
Seed variance	Diffusion restorers	How much the output changes with the random seed alone. High variance means the detail is a sample from a prior.
Cross-model disagreement	Deterministic restorers	Where independent models disagree, at least one is inventing.
Multi-frame agreement	Video sources	The strongest of the four: detail confirmed across independently degraded frames is genuine information gain, not prior-borrowing.
Identity drift	Any track	How far the enhanced crop moved in template space from the original. A restoration that improves the picture and moves the identity is a false lead.

A PRE-REGISTERED GATE THAT HAS NOT YET BEEN PASSED

Adopting any enhancement stage into the *recognition* path requires $\geq +2.0$ points TAR@FAR=0.1% with a bootstrap confidence interval excluding zero, on a real degraded benchmark rather than a synthetically degraded clean one, with no benchmark regressing by more than 0.5 points. This gate was fixed in writing before any run. Investigator *viewing* value and recognition gain are separate claims, and only the second is gated.

SECTION 14

Frontend and investigator workspace

A React single-page application built with Vite, serving two distinct audiences from one origin: a public marketing surface and an authenticated investigator workspace.

Surface	Function
Verify	1:1 comparison with the full result panel: score, decision band, quality reasons, per-stage timings, and what the image itself contains.
Search	1:N identification against the tenant gallery, ranked candidates with runner-up margin, adjudication controls.
Batch compare	One reference against many probes, or pairwise, wired to the real batch endpoint.
Enrol	Gallery management: subjects, templates, per-template deletion.
Cases	The investigative container, with lawful basis captured at creation and carried into every search.
Enhancement	Split view with synchronised zoom, the plan trace including skipped stages and why, metric deltas, and processing history. The original is always beside the output.
Audit	The chain, readable, with an integrity check.

Two things are enforced in the interface rather than left to the operator. Enhanced output is never shown without the original beside it and its label. Adjudication is a recorded act with an actor and a timestamp, not a transient UI state — an examiner's judgement on a candidate is part of the record.

SECTION 15

Reporting and court output

The report is generated from real search results, not from a template with fields filled in. It is required to state:

- The recogniser backend, the model pack, and the pack version actually used.
- Both thresholds in force at the time of the search, and the decision band the score fell into.
- The gallery size searched and the runner-up margin.
- The quality assessment and every reason that set the review flag.
- The evidence layer's output where available, on the standard verbal scale, with its declared reference population named.
- Where enhancement was used: the model, the version, the parameters, the measured synthesis fraction, and the identity drift — with the original image beside the output.
- An explicit limitations section, and the audit chain excerpt for the search.

THE RULE THAT SHAPES THE REPORT

The report must make it possible for a reader to disagree with it. A report that states a conclusion without the operating point, the population, the gallery size and the margin cannot be contested, and a forensic conclusion that cannot be contested is not a forensic conclusion.

SECTION 16

Observability and performance envelope

Every request carries per-stage timing: decode, detect, align, embed, quality, total. These are returned in the response and persisted with the search run, so a latency regression is attributable to a stage rather than to the system.

Measure	Value	Note
Probe encode (end to end)	~14.7 ms	Dominated by the recogniser, not by detection or alignment.
Detection	54 ms → from 939 ms	The rebuild removed a redundant second detection pass.

Measure	Value	Note
Gallery search, 10k templates	1.087 ms p50	About 7% of the encode. Below 10■ the search is not the cost.
Concurrency ceiling	~4 workers	Threading saturates here. Batching is the scaling lever, and this is recorded as a bound rather than hidden.
Within-batch deduplication	11× GPU saving	Realistic batches repeat probes; deduplicating within the batch avoids re-encoding them.
Backbone load	On cache miss only	A warm process does not re-pay it.

SECTION 17

Module reachability discipline

The source tree contains 102 Python modules. **56 are reachable from the running service.** That distinction is computed by walking the import graph from the service entry point, not asserted from a directory listing, and both halves are published.

Category	Count	What it means
Reachable from the service	56	Executed, directly or transitively, on a request. This is the delivered system.
Training and dataset tooling	17	Produces and audits models offline. Never in a request path, by design.
Instrumentation	13	Benchmark and audit scripts run from the command line. Part of the delivery; their outputs are the benchmark record.
Packaging and export	4	Model export and client packaging. Offline.
Superseded	4	Earlier implementations retained in history.
Unreached, no specific role	7	Present, not invoked. Listed rather than quietly removed.
Unclaimed capability	1	Presentation-attack detection. Deliberately not wired in, because the capability is not claimed.

Listing every file as part of the delivered system would let a reader conclude that a capability exists because a file named after it is present. For a forensic system that is exactly the wrong inference to invite. For a maintainer, the unreachable list is the more useful of the two.

PART B

The target architecture

Part B describes where the system is going and why. It is a design document: nothing in it is built except where explicitly marked, and the migration map in Section 22 states what survives the transition.

SECTION 18

Why the current formulation is at its limit

The deployed pipeline is not one pipeline. It is five consecutive irreversible commitments, each of which destroys exactly what forensic inference needs.

Stage	What it destroys	Why forensics needs that
Detection	The scene	Absolute scale, camera geometry, and other observations of the same person in the same frame.
Alignment to 112×112	Imaging metadata	Point-spread function, motion-blur direction, true face size in pixels, sensor response, compression history.
Embedding to a fixed 512 dimensions	Information content	A 1000-pixel portrait and a 19×14 crop emit the same 512 floats. The representation asserts they carry equal information, which is false by many bits.
Cosine similarity	Population structure	How common this face is, who else it resembles, the local density around the candidate.
Threshold	Everything but one bit	Evidential weight, uncertainty, and the possibility of answering 'insufficient'.

An astronomer does not discard the point-spread function before doing photometry; it is the single most important thing they know about the instrument. Stage two of this pipeline discards the equivalent, as preprocessing — and the field then treats degraded imagery as an intractable modelling problem rather than as a **known operator applied in the forward direction**.

SECTION 19

The layered target design

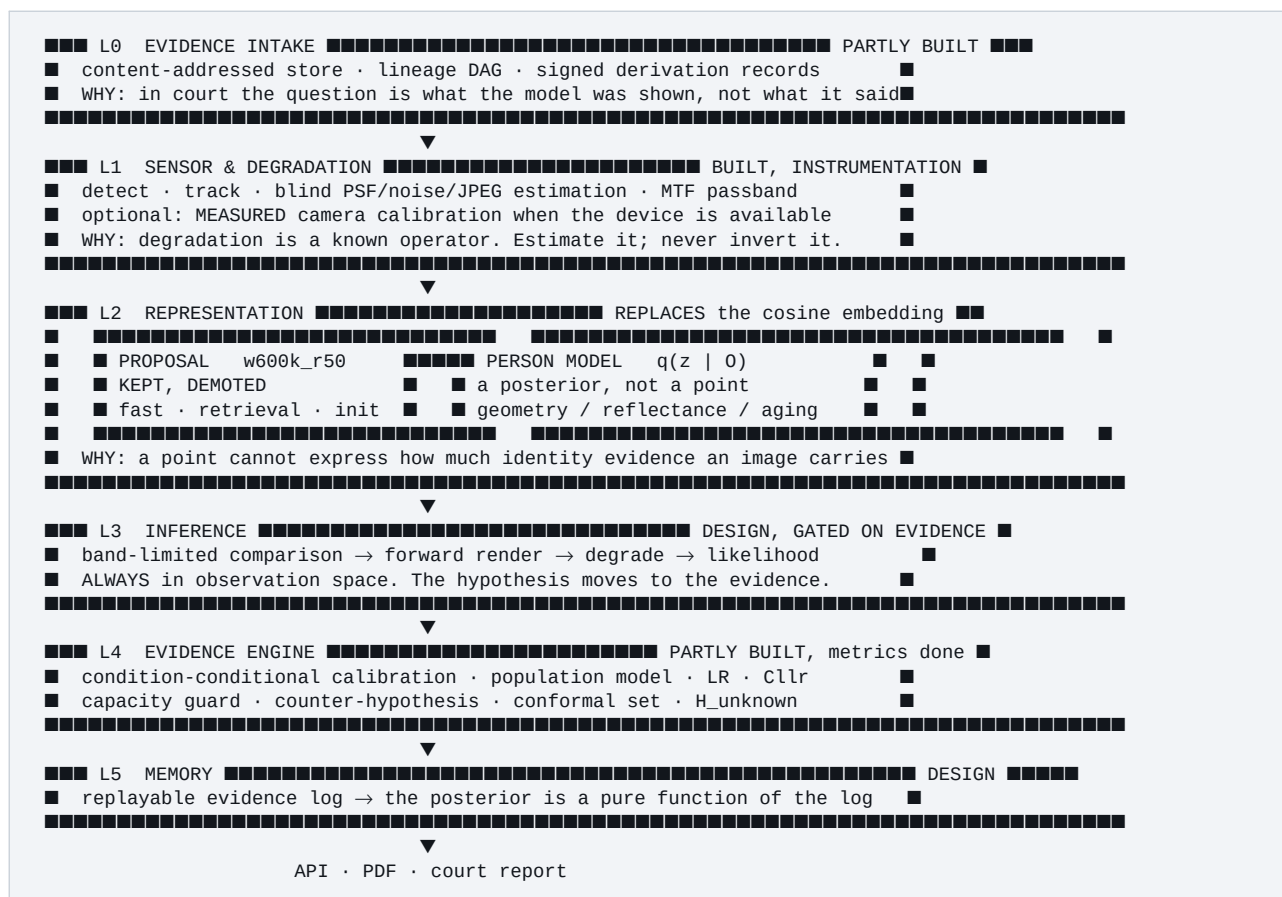


Figure 19.1 — the target layering, each layer marked with its true present state.

SECTION 20

Identity representation

A person is not a vector. In the target design a person is a conditional generative model, and what is currently called an embedding becomes a posterior over its parameters.

Property	Current representation	Target representation
Form	A point in a 512-dimensional metric space	A distribution with support: a posterior over person-model parameters
Information content	Assumed constant regardless of the image	Data-dependent — a 19×14 crop yields a posterior barely narrower than the population prior, which is the truth and can then be reported
Factorisation	Dominated by appearance statistics that are largely nuisance	Factored by physical invariance: craniofacial geometry, reflectance, and an aging flow, with identity concentrated in the near-constant geometry term
Time	Static; cross-age comparison is a distance between endpoints	A trajectory; cross-age comparison is transport along a learned flow
Comparison	Cosine similarity against a threshold	A marginal likelihood ratio against a declared reference cohort

Why the factorisation matters for the target condition

At CCTV resolution reflectance detail is not degraded, it is *absent*. A 20-pixel face carries almost no reflectance information but still constrains coarse geometry. A representation that concentrates identity in the geometry term can express that; a representation dominated by appearance statistics weights the signal roughly inversely to its identity content.

Why this may work now when the same idea failed in 1999

The strongest objection to the whole design is that analysis-by-synthesis was tried with 3D morphable models and deep discriminative learning beat it. That objection deserves a direct answer.

1999 failure cause	Status in 2026
The generative model was a linear decomposition over roughly two hundred laser scans	Diffusion models, Gaussian splatting and neural geometry trained over millions of faces
Optimisation took minutes per image and fell into local minima	An amortised proposal plus differentiable rendering brings it under a second
Rendering was not differentiable	Fully differentiable rasterisation and splatting
The approach could not absorb large data	It now trains at web-scale face corpora

All four causes were engineering, and all four have inverted. The usual reading of the scaling lesson is that structure loses to scale; the honest version is that *structure which cannot absorb scale* loses. This structure now can. That is the bet, and the project's own cheap falsification experiment was designed to decide it before committing years — the companion implementation-plan volume records that it was run, and what it returned.

SECTION 21

The three anti-hallucination invariants

The requirement is absolute: the system must never invent facial detail that is not in the evidence. Three mechanisms, in increasing strength.

1 · Comparison happens in observation space, only

The generative model renders *down* to the evidence. The evidence is never rendered *up*. The system evaluates the likelihood of the observed image under a hypothesis; it never constructs a high-resolution reconstruction and compares embeddings of that. These are not two ways of doing the same thing. Super-resolution is one-to-many and picks a plausible mode from a learned prior, so matching against it is matching against the generator. Forward rendering is many-to-one and well-posed. This is enforced by type-level separation, and the enhancement subsystem's type barrier described in Section 13 is the same discipline already in force today.

2 · An information-theoretic runtime guard

The evidence layer estimates the identity information available in an observation of a given quality. The reported evidential weight is asserted against that bound: a hypothesis cannot be supported more strongly than the pixels can support. **If the generative prior leaks into the conclusion, the reported weight exceeds the channel capacity of the evidence and the guard fires.** This converts an abstract safety property into a runtime assertion.

STATED LIMITATION

The bound is an expectation over comparisons, not a hard per-case limit. It is therefore a strong heuristic guard rather than a proof. It is calibrated to fire conservatively, and every firing is logged rather than silently clamped. Saying this plainly is part of the design; a guard whose limits are not stated will be trusted past them.

3 · A prior-leakage null test, per case

Inference is re-run with the identity term ablated — the renderer conditioned on the population mean rather than on the specific person. If substantial 'evidence' survives that ablation, it came from the prior rather than from the image. This is run per case and reported in the record, not sampled during development and assumed to hold.

SECTION 22

Migration map

Component	Decision	Reason
SCRFD detector	Keep	Measured, adequate, not the bottleneck.
Hash-chained audit	Keep, extend to a lineage graph	Already the right idea; it needs to become the data model rather than a side record.
Forensic metrics	Keep	Built and validated. Load-bearing for everything above.
Threshold configuration	Keep as the fallback path	It must survive as the degraded-mode answer when inference is unavailable.
Exact inner-product search	Keep to 10■, then reconsider	Recall, not speed, decides this — and the trigger is a measured gallery size.
w600k_r50 embedding	Demote from answer to proposal	Fast retrieval and an inference initialiser. It is not the identity representation.
Bespoke quality filter	Replace with the published standard	A standard is verifiable by a procurement officer. A bespoke metric is not.
Cosine plus threshold	Replace	The calibration measurements show the decision layer is not where the loss is. The representation is.
Fixed embedding, image-as-atomic-unit, template store	Replace	Each is a consequence of treating recognition as a memoryless function from images to a metric space.
Scientific content	Discard nothing	Every negative result and every refuted thesis stays in the record.

THE RAREST ASSET, PRESERVED DELIBERATELY

The measurement discipline is the thing that must survive the migration intact: contamination reported as floors rather than totals; a benchmark gap diagnosed as a provenance problem rather than an accuracy problem; a published negative fine-tune; a promising result overturned by its own control. Every module in Part B depends on that discipline being real. An architecture is only as trustworthy as the measurements that justify it.

SECTION 23

Architecture decision record

Decision	Verdict	Evidence
Replace the R50 recogniser with the ViT-B KP-RPE backbone in production	Adopted	One variable changed. TinyFace TAR@FAR=0.1% 20.59% → 60.77%. Clean-set regression checked and accepted. Mandatory consequence recorded: thresholds recalibrated, not carried over.
Condition removal before scoring	Adopted	Embedding-space centering with within-class covariance normalisation, plus score-space margin and per-probe normalisation. Measured gain on the target condition.
Quality-routed model selection for 1:1	Adopted	+4.23 points TinyFace TAR@FAR=0.1% for a 0.13-point worst-case clean cost, with the routing threshold derived on data disjoint from the benchmarks. The only unambiguous model improvement in the project — and it came from routing, not from a bigger model.
Keep 1:N single-model until embedding compatibility is verified at gallery scale	Adopted	Rank ordering is more sensitive to representation drift than a threshold comparison is. Routing a 1:1 decision is safe; routing a ranked list is not, until measured.

Decision	Verdict	Evidence
Multi-frame template averaging	Rejected	Measured, no gain that survived its own control.
Three-model ensemble	Rejected	Lost to the best single model on four of five datasets at three times the compute.
Approximate nearest-neighbour indexing	Deferred on a measured trigger	Recall loss is invisible to the examiner. Adopt on a gallery-size trigger with the loss measured and accepted, not on principle.
Generative restoration in the evidential path	Rejected, structurally	Enforced by the type system rather than by policy. This is the invariant the whole target architecture is built around.
Kinship and population-substructure modelling	Refused	Scientifically interesting; it implicates people who never consented. Not built.

SECTION 24

Known limits

Stated here so they are not discovered later.

- **Surveillance imagery remains the honest limit.** The system delivers a small fraction of a one-bit question on native low-resolution capture, and 98.7% of that shortfall is discrimination loss. Better reporting cannot close it.
- **The population model is the critical path for the evidence layer**, and it is blocked on identity-labelled reference cohorts rather than on code.
- **The forward-operator thesis was refuted on real optics.** Modelling a camera's degradation and applying it forward did not beat simple resolution matching on real distance capture. The layer that depended on it is gated on that result, and the refutation is published.
- **Concurrency saturates at about four workers.** Batching is the scaling lever; more workers is not.
- **Presentation-attack detection is not claimed.** The liveness and synthetic-media signals are passive heuristics, marked uncertified.
- **No independent validation yet.** Every number in this document was produced in-house. External submission is planned and has not happened, and until it does these are self-reported measurements — accurate, protocol-stated, and self-reported.

CLOSING

The durable position this architecture is built toward is not accuracy. It is a system that states its own limits provably, can prove what it concluded without a retracted piece of evidence, and never converts a prior into evidence. A system with those three properties will be trusted in contexts where a more accurate system is not — and that position is not reachable by scaling a matcher.